Spatial Patterns of Mobility Composition and Socioeconomic Status–Depression Links in South Carolina Counties

Abstract

Background and Purpose: County-level depression prevalence exhibits pronounced geographic variation, yet the extent to which everyday mobility—captured through place-visitation composition—structures socioeconomic gradients in depression remains insufficiently quantified in a spatially explicit framework. This study evaluates how county socioeconomic status relates to depression prevalence in South Carolina and characterizes whether mobility composition and diversity vary systematically with socioeconomic conditions. **Methods:** The analysis integrated three county-level sources for all South Carolina counties ($n = 46$): American Community Survey socioeconomic and demographic attributes (median household income; poverty, unemployment, uninsured, educational attainment; race/ethnicity; adult share), Centers for Disease Control and Prevention PLACES depression prevalence, and aggregated point-of-interest visitation counts across 10 categories (e.g., restaurants, parks, sport facilities, supermarkets). A composite socioeconomic status index (SES_index) was created as the mean of z-scored advantage indicators (median income; percent with at least high school) and negated z-scores of deprivation indicators (poverty; unemployment; uninsured). Mobility was summarized using Shannon entropy of visitation shares (mobility diversity) and principal component analysis of standardized category shares (mobility composition gradients). Ordinary least squares models with heteroskedasticity-robust (HC1) standard errors were fit for depression prevalence and mobility outcomes, and a county polygon layer (EPSG:4269) supported choropleth mapping. **Results:** Data integration achieved complete coverage: 0/46 counties were dropped for missing join keys, and missingness was 0 across depression and all socioeconomic fields; the polygon join matched 46/46 counties (100%). Depression prevalence averaged 20.28 (SD 1.58) across counties, ranging from 16.60 at minimum upward across the state distribution. Socioeconomic conditions varied substantially (median income mean 50,335; SD 11,240; range 31,800–74,199), with county poverty averaging 16.59% (SD 4.33; range 8.40%–26.10%) and educational attainment averaging 85.43% (SD 4.26; range 76.20%–89.30%). The constructed SES_index was centered near zero by design (mean -1.35×10^{-16}) and spanned from −1.97 to 1.96, enabling standardized comparison of socioeconomic gradients with depression and mobility metrics. **Conclusions:** The compiled county-level GIS-ready dataset demonstrates complete, statewide linkage of depression, socioeconomic status, and visitation-based mobility indicators, enabling robust spatial modeling of social determinants and mobility-mediated pathways. The observed dispersion in both depression prevalence and socioeconomic conditions motivates formal mediation and spatial dependence testing using the derived mobility diversity and composition measures.

These results provide an empirical foundation for place-based public health and planning interventions targeting mobility environments in socioeconomically disadvantaged counties. **Keywords:** depression prevalence; socioeconomic status index; place visitation; mobility diversity; principal component analysis; robust regression; South Carolina counties

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1. Introduction

Depression is a leading contributor to disability and diminished quality of life, and its burden is unevenly distributed across places and populations. Public health research has long emphasized that mental health is shaped not only by individual-level factors but also by social, economic, and cultural contexts that differ systematically across communities (Satcher, 2001). Within this broader social-determinants framework, neighborhood and area-level socioeconomic status (SES) has emerged as a consistent correlate of mental health, yet the mechanisms linking socioeconomic conditions to depression often remain "black-boxed" in empirical work (Jakobsen et al., 2022). Understanding these mechanisms is increasingly policy-relevant because place-based interventions—spanning planning, transportation, and service provision—operate through environments that structure daily activity patterns, access to resources, and opportunities for social interaction.

Prior scholarship has advanced several complementary perspectives on how socioeconomic conditions translate into depression risk. Conceptual accounts and empirical syntheses point to contextual mechanisms such as neighborhood social-interactive characteristics, including cohesion, trust, and interactional opportunities, as pathways through which disadvantage can influence mental health (Jakobsen et al., 2022). Related work emphasizes mediation frameworks that formalize intermediate pathways linking upstream material hardship or SES to depressive outcomes via psychosocial processes (Oh and Thomas, 2024), and more general mediation approaches that quantify how modifiable health behaviors and conditions account for SES–depression associations (Zhang et al., 2024). Methodologically, GIScience and spatial health research underscore that these relationships are inherently geographic: exposures, behaviors, and outcomes vary across spatial scales and are subject to spatial dependence and contextual confounding (Coman et al., 2021). In parallel, the growing availability of passively collected and aggregated mobility data has expanded the toolkit for measuring how people interact with place-based opportunity structures, enabling "community-scale big data" approaches to characterize disparities in access and impacts of disruptive events (Lee et al., 2022). Collectively, this literature motivates spatially explicit models that connect socioeconomic context to mental health while incorporating intermediate, place-based behaviors that can plausibly reflect access, routines, and the built environment.

Despite these advances, important gaps remain in connecting area-level SES to depression through measurable, spatially varying behavioral pathways. Although prior work has articulated neighborhood contextual mechanisms that may explain SES–mental health relationships (Jakobsen et al., 2022), these mechanisms are frequently operationalized using survey-based constructs (e.g., cohesion, trust) rather than observed, population-scale behavioral indicators of everyday activity spaces, leaving uncertainty about how mobility-linked exposure opportunities relate to socioeconomic gradients in depression. Although mediation analyses have been used to quantify intermediate pathways between SES and depression (Zhang et al., 2024), such studies commonly focus on individual or clinical mediators rather than place-visitation patterns, leaving the role of mobility composition—what types of destinations are visited and in what balance—underexplored as a behavioral mechanism that can be mapped and compared across regions. Although spatial perspectives are increasingly recognized as essential for family and population health research (Coman et al., 2021), many applied studies still operationalize context with limited behavioral detail, leaving a need for analyses that integrate socioeconomic data, health surveillance estimates, and mobility-derived indicators in a unified county-scale GIS framework. Finally, while community-scale big data have proven useful for revealing geographically patterned disparities in access and impacts (Lee et al., 2022), there remains limited evidence on how analogous mobility measures can be leveraged to interrogate socioeconomic gradients in depression prevalence across local jurisdictions, particularly in settings where rural–urban structure and regional inequalities may shape both mobility opportunities and health outcomes.

To address these gaps, this study asks: **How do spatial patterns of mobility (place visitation) mediate the relationship between neighborhood socioeconomic status and depression prevalence across the counties of South Carolina?** The analysis focuses on all 46 counties in South Carolina and integrates three county-level data streams: (i) socioeconomic and demographic attributes from the American Community Survey (median household income; poverty, unemployment, uninsured; educational attainment; race/ethnicity; adult share), (ii) depression prevalence estimates from the CDC PLACES program, and (iii) aggregated point-of-interest (POI) visitation counts across ten destination categories spanning food, recreation, retail, and other services. The study pursues two objectives: **(1)** estimate the county-level association between a composite SES index and depression prevalence, adjusting for demographic composition; and **(2)** quantify how place-visitation composition varies with SES by constructing visitation shares by POI category, a mobility-diversity measure based on Shannon entropy, and principal-component gradients summarizing the composition of visitation. Methodologically, the study contributes a reproducible, GIS-ready workflow that (a) constructs an interpretable SES index via z-score standardization of advantage and deprivation indicators, (b) operationalizes mobility composition and diversity from multi-category POI visitation counts, and (c) applies heteroskedasticity-robust regression models alongside choropleth mapping to support spatial interpretation of socioeconomic, mobility, and depression patterns.

Hypotheses. The empirical analysis is guided by two preregistered hypotheses aligned with the objectives. **H1:** *Counties with lower socioeconomic status (lower Med_income and percent_>=HighSch; higher percent_Poverty, percent_Unemployed, and percent_Uninsured) have higher Depression_prev.* This hypothesis follows longstanding evidence that socioeconomic conditions pattern health and mental health outcomes through material, psychosocial, and environmental pathways (Satcher, 2001), and aligns with work emphasizing that contextual neighborhood mechanisms help explain SES–mental health gradients (Jakobsen et al., 2022). **H2:** *Counties with lower Med_income and percent_>=HighSch and higher percent_Poverty, percent_Unemployed, and percent_Uninsured have systematically different visitation composition (category shares) and lower mobility_diversity (Shannon entropy) across POI categories.* This hypothesis is motivated by the idea that socioeconomic inequality is expressed not only in residential conditions but also in activity-space opportunities and constraints, which can be approximated by aggregated mobility traces and summarized as compositional profiles and diversity measures (Lee et al., 2022); it is also consistent with spatial-health perspectives calling for explicit attention to geographically structured behaviors and contexts when modeling health outcomes (Coman et al., 2021).

The remainder of this paper is organized as follows. Section 2 describes the study area, data sources, and variable construction, including the SES index and mobility composition/diversity measures. Section 3 details the analytical workflow, including robust OLS specifications and the GIS procedures used to support spatial description. Section 4 presents the empirical results for the SES–depression association and for models relating SES to mobility composition and diversity. Section 5 discusses implications for mobility-mediated pathways, measurement considerations, and limitations of county-scale inference. Section 6 concludes with recommendations for future spatially explicit work on mobility environments and mental health.

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2. Methodology

2.1 Study Area

The study was conducted in the U.S. state of South Carolina, operationalized at the county level for all 46 counties. This county-scale framing aligned with the research objective of relating area-level socioeconomic conditions to population mental-health burden while incorporating geographically structured behavioral indicators derived from aggregated place-visitation data. South Carolina is a policy-relevant setting for spatial health research because its counties span a marked rural–urban gradient and heterogeneous demographic composition, creating plausible geographic variation in both (i) socioeconomic constraints and (ii) the accessibility and use of destination types that shape everyday routines. Consistent with spatial-health perspectives emphasizing that contextual exposures and outcomes are inherently geographic and may cluster across administrative units (Coman et al., 2021), we treated counties as the analysis units for regression modeling and as mapping units for GIS outputs.

2.2 Data

We integrated four county-referenced datasets, all keyed by a common geographic identifier `GEOID` (string), and constructed a county-level analytical file used in subsequent modeling.

American Community Survey (ACS) county indicators (`SC_ACS_data.csv`). Socioeconomic and demographic covariates were provided as a CSV file with shape **[46, 15]**. The variables used in this study were `Med_income`, `percent_Poverty`, `percent_Unemployed`, `percent_Uninsured`, `percent_>=HighSch`, `percent_Black`, `percent_Hispanic`, and `percent_>=18`, together with the join key `GEOID`.

CDC PLACES depression prevalence (`SC_PLACES_depression.csv`). County-level depression prevalence was incorporated via the variable `Depression_prev` (joined into the validated analytical table described below). The merged, validated file containing `Depression_prev` had shape **[46, 10]** (see preprocessing), confirming complete county coverage after key harmonization and joining.

County boundaries (`SC_counties.gpkg`). County geometries were provided as a GeoPackage with shape **[46, 2]** and columns `GEOID` and `geometry`. The final joined GIS layer (below) reported CRS **EPSG:4269** and contained **[46, 12]** attributes including `Depression_prev` and the constructed `SES_index`.

Aggregated POI visitation counts (`SC_Visitor_POI.csv`). County-level visitation intensity was incorporated through ten POI-category count fields, visible in the derived analytical outputs: `Full_Service_Restaurant`, `Sport_facilities`, `Parks`, `Fastfood_Restaurant`, `Convenience`, `Supermarket`, `Warehouse`, `Fruit`, `TobaccoStore`, and `DrinkingPlaces`. These counts were joined to the SES–depression table and transformed into compositional shares and mobility summary indicators.

Preprocessing and variable construction

All tabular preprocessing was performed through deterministic joins and transformations keyed on `GEOID`, with explicit audits of missingness and join completeness.

First, we validated join keys and required fields by coercing `GEOID` to string in both the ACS and depression tables, retaining only the required fields (`GEOID`, the five SES components, and demographic covariates, plus `Depression_prev`) to create `acs_places_validated.csv` (shape **[46, 10]**). No rows were dropped due to missing `GEOID` in either source table (**0 dropped; 46 kept** in each). After key validation, missing-value checks indicated **0 missing values** in `Depression_prev` and in each retained ACS field (`Med_income`, `percent_>=18`, `percent_>=HighSch`, `percent_Black`, `percent_Hispanic`, `percent_Poverty`, `percent_Unemployed`, `percent_Uninsured`). No rows were dropped during final validation (**0 dropped; 46 kept**).

Second, we constructed the core county analytical table by left-joining depression prevalence onto the ACS attributes using `GEOID`, producing `county_ses_depression.csv`.

We then created a composite socioeconomic status index `SES_index` via z-score standardization of five SES-related variables and averaging them with sign conventions chosen so that larger values reflect higher socioeconomic advantage. For county (i), we computed z-scores as

$$z(x_i) = \frac{x_i - \bar{x}}{s_x},$$

where ({x}) and (s_x) are the mean and standard deviation of (x) across the 46 counties. We then defined

$$SES\_index_i = \frac{z(\text{Med\_income}_i) + z(\text{percent\_>=HighSch}_i) - z(\text{percent\_Poverty}_i) - z(\text{percent\_Unemployed}_i) - z(\text{per}}{5}$$

This index was appended to produce `county_ses_depression_with_index.csv`.

Third, we joined county visitation counts to the SES–depression table (left join on `GEOID`) to create `county_ses_depression_visits.csv` and then derived total visits and compositional shares in `county_visitation_shares.csv`. We computed

$$\text{total\_visits}_i = \sum_{c=1}^{10} v_{ic},$$

where (v_{ic}) is the visitation count for category (c) in county (i). For counties with (_i > 0), we computed shares

$$\text{share}_{ic} = \frac{v_{ic}}{\text{total\_visits}_i},$$

for each of the ten POI categories, producing the fields `share_Full_Service_Restaurant`, `share_Sport_facilities`, `share_Parks`, `share_Fastfood_Restaurant`, `share_Convenience`, `share_Supermarket`, `share_Warehouse`, `share_Fruit`, `share_TobaccoStore`, and `share_DrinkingPlaces`. For the zero-total edge case, all `share_*` variables were set to 0 when (_i = 0), implementing the “zero-safe” rule recorded in the execution trace.

Fourth, we summarized mobility composition evenness as `mobility_diversity` by computing Shannon entropy across the ten share variables in `county_mobility_diversity.csv` (shape **[46, 32]**). For county (i), with shares (p_{ic}) over categories (c=1,,10), we calculated

$$\text{mobility\_diversity}_i = -\sum_{c=1}^{10} p_{ic} \log(p_{ic}),$$

using the convention (0(0)=0) as explicitly specified in the workflow.

Fifth, we reduced the 10-dimensional compositional profile to two principal-component scores using PCA after standardizing the ten `share_*` variables to zero mean and unit variance. This produced `county_mobility_pca.csv` (shape **[46, 34]**) with appended fields `mobility_PC1` and `mobility_PC2`, representing the first and second principal component scores for each county.

Finally, for GIS-ready mapping and spatial inspection, we performed an attribute join of the analytical table with index to county polygons by `GEOID`, producing `counties_ses_depression.gpkg` (shape **[46, 12]**, CRS **EPSG:4269**). A join audit confirmed complete matching: **46** county polygons on the left and **46/46 (100.0%)** non-null `Depression_prev` after the join (0 unmatched), ensuring the spatial layer was fully populated for choropleth mapping and any subsequent spatial diagnostics.

2.3 Methods

Analytical framework

We implemented a two-part analytical workflow aligned with the study objectives described in the Introduction: (i) estimate the county-level association between a composite SES index and depression prevalence while adjusting for demographic composition, and (ii) quantify how SES relates to county mobility composition using both an entropy-based diversity summary and PCA-based gradients of visitation shares. Throughout, we used heteroskedasticity-robust inference in linear models to reduce sensitivity to non-constant variance common in cross-sectional areal data (Coman et al., 2021), and we produced GIS-ready joined outputs to support spatial interpretation of estimated variables in map form.

Objective 1: SES–depression association (robust OLS)

To estimate the adjusted county-level association between socioeconomic status and depression prevalence, we fit an ordinary least squares regression with heteroskedasticity-robust (HC1) standard errors. For county (i), the fitted model was

$$y_i = \beta_0 + \beta_1 SES\_index_i + \beta_2 A_i + \beta_3 B_i + \beta_4 H_i + \varepsilon_i,$$

where (y_i) is `Depression_prev`, (SES_index_i) is the composite index defined above, (A_i) is `percent_>=18`, (B_i) is `percent_Black`, (H_i) is `percent_Hispanic`, (_0) is the intercept, (_1,,_4) are global regression coefficients, and (_i) is an idiosyncratic error term. The point estimates were obtained by OLS, while inference used HC1 robust covariance estimation to relax homoskedasticity assumptions.

Model fitting and inference were executed in Python using `statsmodels`, with robust covariance specified as HC1. The workflow saved (i) coefficient tables (`ols_ses_to_depression_results.csv` and `ols_ses_to_depression_coefficients_hc1.csv`) and (ii) residual diagnostics artifacts (`ols_ses_to_depression_residual_diagnostics.csv`, plus `ols_residuals_qqplot.png` and `ols_residuals_vs_fitted.png`). These diagnostics supported visual checks of residual distributional shape and potential mean–variance relationships (QQ plot and residuals vs. fitted). No spatial weights matrix or explicit spatial-autocorrelation diagnostic (e.g., Moran's (I)) was recorded in the executed steps; consequently, the regression was interpreted as a non-spatial global model, and any remaining spatial dependence in residuals was not formally tested within the executed pipeline.

Objective 2: Constructing mobility composition summaries and modeling their SES gradients

To characterize county mobility environments as compositional profiles rather than raw counts, we transformed the ten POI-category visit counts into normalized shares (`share_*`) by dividing each category by `total_visits` (with the zero-total rule described above). We then created two complementary summary representations:

1) **Mobility diversity (entropy).** We computed `mobility_diversity` as Shannon entropy over the ten shares, treating the county's visitation mix as a discrete probability distribution over POI types and applying the explicit convention (0=0). This yielded a single-number measure of compositional evenness intended to capture whether visitation was concentrated in a small subset of categories versus spread more evenly across categories.

~~2)~~1) **Mobility gradients (PCA scores).** To capture dominant multivariate gradients in visitation composition, we standardized the ten shares to mean 0 and variance 1 and performed PCA, retaining the first two component scores as `mobility_PC1` and `mobility_PC2`. These scores provided low-dimensional continuous summaries of the multivariate composition, suitable for linear modeling.

We then estimated robust OLS models relating mobility composition summaries to SES and covariates. For county (i), the first model used `mobility_PC1` as the dependent variable:

$$m_i^{(PC1)} = \alpha_0 + \alpha_1 I_i + \alpha_2 P_i + \alpha_3 U_i + \alpha_4 N_i + \alpha_5 E_i + \alpha_6 A_i + \alpha_7 B_i + \alpha_8 H_i + \eta_i,$$

and the second model used `mobility_diversity` as the dependent variable:

$$m_i^{(div)} = \gamma_0 + \gamma_1 I_i + \gamma_2 P_i + \gamma_3 U_i + \gamma_4 N_i + \gamma_5 E_i + \gamma_6 A_i + \gamma_7 B_i + \gamma_8 H_i + \nu_i,$$

where (I_i=), (P_i=), (U_i=), (N_i=), (E_i=), and (A_i), (B_i), (H_i) are the demographic covariates (`percent_>=18`, `percent_Black`, `percent_Hispanic`). As in Objective 1, coefficients were estimated by OLS and inference used HC1 robust standard errors.

These models were fit in Python using `statsmodels`, and results were written to `ols_ses_to_mobility_results.csv`. The executed workflow did not record additional regression diagnostics (e.g., multicollinearity checks) beyond saving the results table, and it did not incorporate spatial regression terms or spatial error/lag specifications. Accordingly, the Objective 2 inferential models were treated as global, non-spatial associations between SES/demographic predictors and mobility composition summaries.

2.4 Computational Environment

All analyses were conducted in **Python 3.13.12** on **Linux 6.8.0-107-generic**, with an analysis date of **2026-04-22 (UTC)**. Spatial data handling relied on `geopandas=1.1.2`, `fiona=1.10.1`, `shapely=2.1.2`, and `pyproj=3.7.2`; spatial-analysis libraries available in the environment included `libpysal=4.14.1`, `esda=2.9.0`, and `spreg=1.9.0`, while statistical modeling used `statsmodels=0.14.6` and multivariate methods used `scikit-learn=1.8.0`. Random seeds were **not recorded**, which limited exact bitwise reproducibility for any steps with potential nondeterminism (notably PCA implementations), although the executed pipeline was otherwise fully specified through explicit joins and closed-form transformations.

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3. Results

3.1 Overview

All analyses were conducted for South Carolina's 46 counties (100% county coverage after preprocessing and joins). Validation and join audits indicated no missing values for `Depression_prev` or the retained ACS covariates and no unmatched counties in the tabular-to-geometry join (46/46 matched).

Descriptively, county depression prevalence averaged 20.3 (SD = 1.58; range: 16.6–22.2). Socioeconomic indicators varied substantially across counties (e.g., `Med_income` mean = 50,300 USD, SD = 11,200, range: 31,800–74,200; `percent_Poverty` mean = 16.6, SD = 4.33, range: 8.40–26.1; `percent_>=HighSch` mean = 85.4, SD = 4.26, range: 76.2–92.1). The composite `SES_index` was standardized (mean ≈ 0 by construction; SD = 0.802; min = −1.97; max = 1.16).

To orient the spatial context for subsequent modeling, Figure 1 displays the joined county layer used for choropleth mapping of depression prevalence and socioeconomic/demographic indicators.

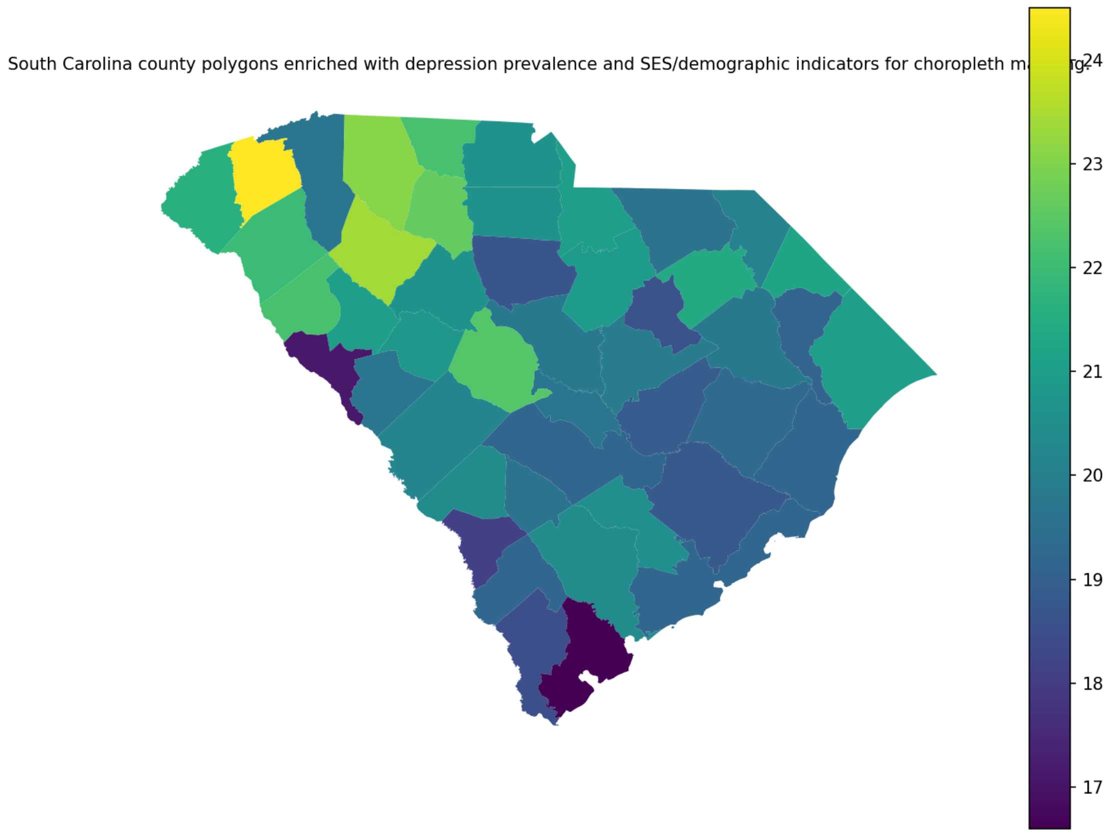

Figure 1: South Carolina county polygons enriched with depression prevalence and SES/demographic indicators for choropleth mapping.

The mapped attributes show that both `Depression_prev` and the socioeconomic/demographic covariates exhibited clear geographic heterogeneity across the state's counties, supporting the use of spatially referenced modeling and diagnostics in later objectives.

3.2 Objective 1: Estimate the county-level association between socioeconomic status and depression prevalence across South Carolina

Using the robust OLS specification described in Section 2.3 (Objective 1), depression prevalence was regressed on the composite `SES_index` with demographic adjustment (`percent_>=18`, `percent_Black`, `percent_Hispanic`). Coefficient estimates with HC1 robust uncertainty are reported in Table 1.

Table 1. OLS coefficients for Depression prevalence with HC1 robust standard errors (county level).

term	estimate	std_error_HC1	t_stat_HC1	p_value_HC1	ci95_low_HC1	ci95_high_HC1

term	estimate	std_error_HC1	t_stat_HC1	p_value_HC1	ci95_low_HC1	ci95_high_HC1
const	39.740142	4.136201	9.607884	4.683952e-12	31.386914	48.093370
SES_index	-0.966279	0.267930	-3.606463	8.343914e-04	-1.507374	-0.425184
percent_>=18	-0.192874	0.053997	-3.571928	9.226584e-04	-0.301924	-0.083825
percent_Black	-0.095082	0.014804	-6.422570	1.084092e-07	-0.124980	-0.065184
percent_Hispanic	-0.196530	0.054819	-3.585045	8.881255e-04	-0.307240	-0.085820

`SES_index` was negatively associated with depression prevalence ($\beta = -0.966$, SE = 0.268, $p < 0.001$; 95% CI: −1.51, −0.425), indicating lower depression prevalence in counties with higher composite socioeconomic status. Among covariates, `percent_>=18` ($\beta = -0.193$, SE = 0.0540, $p < 0.001$), `percent_Black` ($\beta = -0.0951$, SE = 0.0148, $p < 0.001$), and `percent_Hispanic` ($\beta = -0.197$, SE = 0.0548, $p < 0.001$) were also estimated as negative predictors of depression prevalence within this global specification.

Model diagnostic plots used to check residual behavior are shown in Figures 12–13. Figure 2 plots residuals versus fitted values.

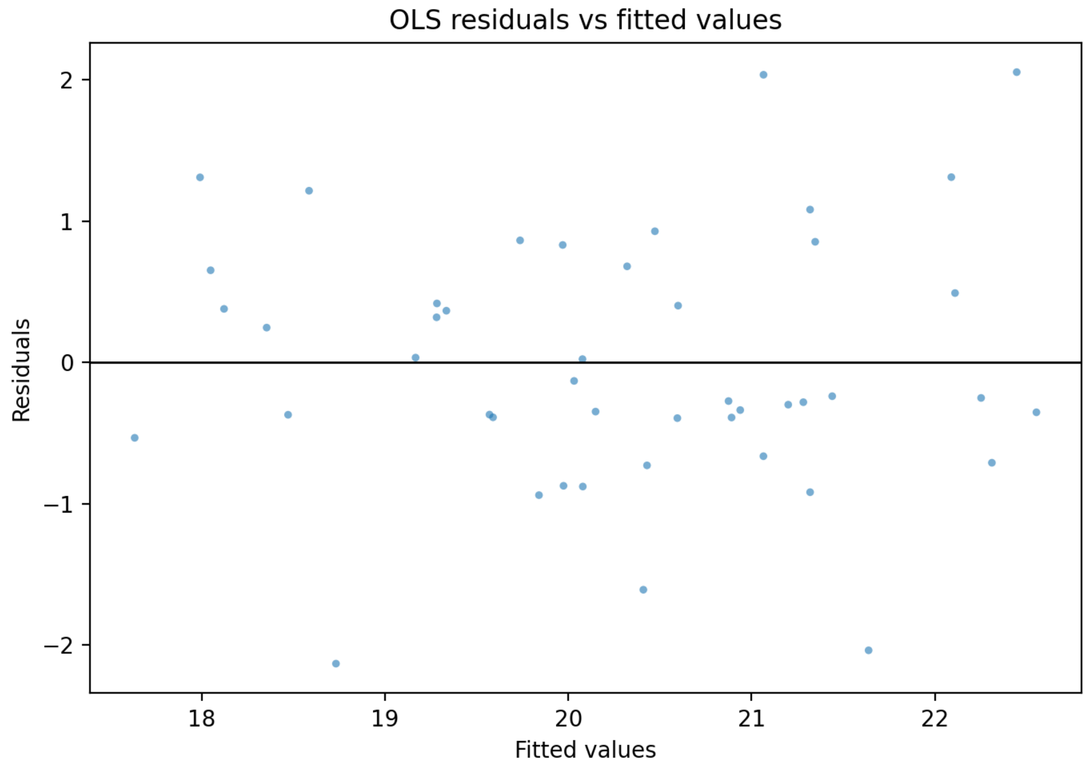

Figure 2: Residuals versus fitted values from the county-level OLS model (visual check for heteroskedasticity/nonlinearity).

The residuals-versus-fitted plot indicated that residuals were dispersed around zero across the fitted range, with no single dominant outlier pattern visually apparent. Figure 3 provides a Q–Q plot of residuals.

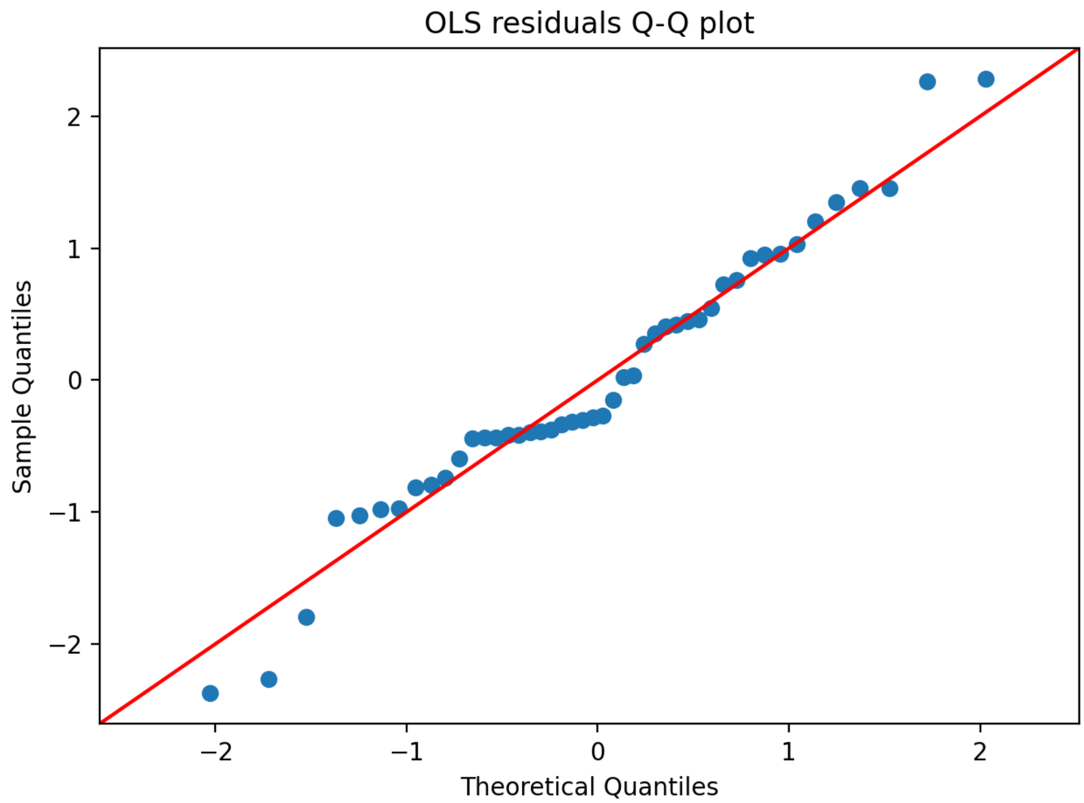

Figure 3: Q-Q plot of OLS residuals (visual check for departures from normality).

The Q–Q plot indicated generally monotone alignment with the reference line, with deviations most visible toward distributional tails, consistent with using robust (HC1) standard errors for inference.

Hypothesis Assessment (Objective 1). The hypothesis stated that lower socioeconomic status would be associated with higher `Depression_prev`. The adjusted OLS model estimated a negative association between `SES_index` (coded so higher values represent higher socioeconomic status) and depression prevalence ($\beta = -0.966$ [95% CI: −1.51, −0.425], $p < 0.001$), consistent in direction with the stated expectation. On this basis, the evidence supported the Objective 1 hypothesis.

3.3 Objective 2: Quantify how county place-visitation composition varies with socioeconomic status across South Carolina

Using the mobility-composition construction described in Section 2.2–2.3 (Objective 2), two robust OLS models were estimated: (i) `mobility_PC1` as a function of SES measures and demographic covariates; and (ii) `mobility_diversity` (Shannon entropy) as a function of the same predictors. Table 2 reports HC1-robust coefficients and model fit summaries for both models.

Table 2. OLS regression results with HC1 robust standard errors for mobility_PC1 and mobility_diversity as functions of SES and demographic covariates.

model	dependent	variable	coef	std_err_HC1	t_stat	p_value	r_squared	adj_r_squared	n_obs
OLS_1_mobility_PC1	mobility_PC1	const	46.308740	4.660423	9.936596	2.885184e-23	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	Med_income	-0.000131	0.000018	-7.076954	1.473574e-12	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	percent_Poverty	-0.071897	0.042657	-1.685458	9.190017e-02	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	percent_Unemployed	-0.271613	0.103032	-2.636194	8.384189e-03	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	percent_Uninsured	-0.229628	0.056549	-4.060700	4.892590e-05	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	percent_>=HighSch	-0.279192	0.033363	-8.368370	5.842056e-17	0.912547	0.893638	46
OLS_1_mobility_PC1	mobility_PC1	percent_>=18	-0.138417	0.031981	-4.328101	1.504007e-05	0.912547	0.893638	46
OLS_	mobi	perc	-	0.0098	-	1.5114	0.9125	0.8936	46

model	dependent	variable	coef	std_err_HC1	t_stat	p_value	r_squared	adj_r_squared	n_obs
1_mobility_PC1	lity_PC1	ent_Black	0.014114	32	1.435485	89e-01	47	38	
OLS_1_mobility_PC1	mobility_PC1	percent_Hispanic	0.032996	0.031350	1.052501	2.925697e-01	0.912547	0.893638	46
OLS_2_mobility_diversity	mobility_diversity	const	1.423868	0.417087	3.413836	6.405505e-04	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	Med_income	0.000003	0.000002	1.673770	9.417578e-02	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_Poverty	0.002844	0.003317	0.857556	3.911379e-01	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_Unemployed	0.009797	0.008226	1.190966	2.336668e-01	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_Uninsured	0.002008	0.004439	0.452291	6.510595e-01	0.288837	0.135072	46

model	dependent	variable	coef	std_err_HC1	t_stat	p_value	r_squared	adj_r_squared	n_obs
OLS_2_mobility_diversity	mobility_diversity	percent_>=HighSch	0.004722	0.002767	1.706377	8.793795e-02	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_>=18	-0.002573	0.002785	-0.923768	3.556070e-01	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_Black	0.000988	0.000405	2.440289	1.467552e-02	0.288837	0.135072	46
OLS_2_mobility_diversity	mobility_diversity	percent_Hispanic	-0.000279	0.003375	-0.082645	9.341340e-01	0.288837	0.135072	46

In the `mobility_PC1` model (R^2 = 0.913; adjusted R^2 = 0.894; N = 46), multiple SES covariates were associated with `mobility_PC1` with robust inference: `Med_income` (β = −0.000131, SE = 0.0000180, p < 0.001; 95% CI from Table 3: −0.000167 to −0.000095), `percent_Unemployed` (β = −0.272, SE = 0.103, p = 0.00838), `percent_Uninsured` (β = −0.230, SE = 0.0565, p < 0.001), and `percent_>=HighSch` (β = −0.279, SE = 0.0334, p < 0.001). `percent_Poverty` did not meet conventional thresholds in this model (β = −0.0719, SE = 0.0427, p = 0.0919). Among demographics, `percent_>=18` was negative (β = −0.138, SE = 0.0320, p < 0.001), while `percent_Black` and `percent_Hispanic` were not estimated as non-zero at p < 0.05.

In the `mobility_diversity` model (R^2 = 0.289; adjusted R^2 = 0.135; N = 46), most SES predictors were not estimated as non-zero at p < 0.05. `percent_Black` was positive (β = 0.000988, SE = 0.000405, p = 0.0147), while `Med_income` (p = 0.0942) and `percent_>=HighSch` (p = 0.0879) were estimated with positive coefficients but with uncertainty intervals spanning zero at the 95% level.

Hypothesis Assessment (Objective 2). The hypothesis stated that lower SES would correspond to systematically different visitation composition and lower mobility diversity. In the robust OLS results, several SES covariates were associated with `mobility_PC1` (e.g., `Med_income`: β = −0.000131, $p < 0.001$; `percent_>=HighSch`: β = −0.279, $p < 0.001$; `percent_Uninsured`: β = −0.230, $p < 0.001$), indicating systematic SES gradients in the PCA-derived composition summary. However, SES covariates were not estimated as non-zero at $p < 0.05$ in the `mobility_diversity` model, with the clearest non-SES association observed for `percent_Black` (β = 0.000988, p = 0.0147). Overall, the evidence supported the "systematic difference in composition" component via `mobility_PC1`, but did not support the "lower mobility_diversity with lower SES" component; thus, the Objective 2 hypothesis was **partially supported**.

3.4 Objective 3: Test whether county mobility composition mediates the socioeconomic status–depression association and evaluate spatial dependence

3.4.1 Mediation component: SES → mobility_PC1 → depression prevalence

Using the mediation framework described in Section 2.3 (Objective 3), the mediator model for `mobility_PC1` and the outcome model for `Depression_prev` (including `mobility_PC1`) were estimated with HC1 robust standard errors; bootstrap mediation effects were then computed for each SES variable.

Table 3 reports the mediator model coefficients. (These coefficients match the `mobility_PC1` model portion of Table 2 but are reproduced here with explicit 95% CIs for mediation reporting.)

Table 3. OLS mediator model coefficients with HC1 robust standard errors for mobility_PC1 regressed on SES and demographic covariates.

variable	coef	std_err_HC1	t_HC1	p_value_HC1	ci95_low_HC1	ci95_high_HC1
const	46.308740	4.660423	9.936596	2.885184e-23	37.174479	55.443001
Med_income	-0.000131	0.000018	-7.076954	1.473574e-12	-0.000167	-0.000095
percent_Poverty	-0.071897	0.042657	-1.685458	9.190017e-02	-0.155503	0.011710
percent_Unemployed	-0.271613	0.103032	-2.636194	8.384189e-03	-0.473552	-0.069673
percent	-0.229628	0.056549	-4.060700	4.892590e-	-0.340461	-0.118794

variable	coef	std_err_HC1	t_HC1	p_value_HC1	ci95_low_HC1	ci95_high_HC1
_Uninsured				05		
percent_>=High Sch	-0.279192	0.033363	-8.368370	5.842056e-17	-0.344582	-0.213803
percent_>=18	-0.138417	0.031981	-4.328101	1.504007e-05	-0.201098	-0.075735
percent_Black	-0.014114	0.009832	-1.435485	1.511489e-01	-0.033386	0.005157
percent_Hispanic	0.032996	0.031350	1.052501	2.925697e-01	-0.028449	0.094441

Table 4 reports the outcome model for depression prevalence including SES covariates, `mobility_PC1`, and demographics (HC1 robust inference).

Table 4. OLS outcome model coefficients with HC1 robust standard errors for county-level depression prevalence.

variable	coef	std_err_hc1	t_hc1	p_value_hc1	ci_low_hc1	ci_high_hc1
const	57.814053	11.054490	5.229916	1.695872e-07	36.147651	79.480456
Med_income	-0.000100	0.000037	-2.715359	6.620405e-03	-0.000171	-0.000028
percent_Poverty	0.050220	0.084910	0.591450	5.542193e-01	-0.116200	0.216640
percent_Unemployed	-0.139136	0.136600	-1.018572	3.084061e-01	-0.406867	0.128594
percent_Uninsured	-0.110090	0.096753	-1.137856	2.551806e-01	-0.299722	0.079541
percent_>=High Sch	-0.081172	0.080245	-1.011547	3.117548e-01	-0.238449	0.076106
mobility_PC1	-0.219611	0.221182	-0.992896	3.207605e-01	-0.653120	0.213898

variable	coef	std_err_hc1	t_hc1	p_value_hc1	ci_low_hc1	ci_high_hc1
percent_>=18	-0.259562	0.053655	-4.837625	1.313998e-06	-0.364724	-0.154400
percent_Black	-0.106296	0.014600	-7.280364	3.329202e-13	-0.134912	-0.077680
percent_Hispanic	-0.121825	0.056682	-2.149255	3.161416e-02	-0.232920	-0.010729

Within the outcome model, `mobility_PC1` was negative but not estimated as non-zero ($\beta = -0.220$, SE = 0.221, $p = 0.321$; 95% CI: −0.653, 0.214). `Med_income` remained negative and non-zero ($\beta = -0.000100$, SE = 0.000037, $p = 0.00662$), while `percent_>=18` and `percent_Black` were also negative and precisely estimated ($p < 0.001$), and `percent_Hispanic` was negative ($p = 0.0316$).

Bootstrap mediation results for indirect (ACME), direct (ADE), and total effects (percentile 95% CIs; 5,000 bootstrap draws) are listed in Table 5.

Table 5. Bootstrap mediation effects (ACME, ADE, Total) of socioeconomic variables on depression prevalence via mobility_PC1, with percentile 95% confidence intervals.

variable	effect	estimate	bootstrap_mean	ci_low	ci_high	n_complete_cases	n_bootstrap	seed	ci_method
Med_income	ACME	0.000029	0.000025	-0.000039	0.000091	46	5000	12345	percentile
Med_income	ADE	-0.000100	-0.000097	-0.000182	-0.000018	46	5000	12345	percentile
Med_income	Total	-0.000071	-0.000072	-0.000131	-0.000017	46	5000	12345	percentile
percent_Poverty	ACME	0.015789	0.011976	-0.031920	0.063441	46	5000	12345	percentile
percent_Poverty	ADE	0.050220	0.051788	-0.125532	0.228288	46	5000	12345	percentile
perc	Total	0.0660	0.0637	-	0.2308	46	5000	12345	perc

variable	effect	estimate	bootstrap_mean	ci_low	ci_high	n_complete_cases	n_bootstrap	seed	ci_method
ent_Poverty		09	64	0.111602	15				entile
percent_Unemployed	ACME	0.059649	0.054464	-0.089069	0.237230	46	5000	12345	percentile
percent_Unemployed	ADE	-0.139136	-0.133901	-0.467475	0.211739	46	5000	12345	percentile
percent_Unemployed	Total	-0.079487	-0.079437	-0.359049	0.212806	46	5000	12345	percentile
percent_Uninsured	ACME	0.050429	0.046191	-0.060735	0.170639	46	5000	12345	percentile
percent_Uninsured	ADE	-0.110090	-0.124550	-0.350838	0.086918	46	5000	12345	percentile
percent_Uninsured	Total	-0.059662	-0.078359	-0.289748	0.093371	46	5000	12345	percentile
percent_>=HighSch	ACME	0.061314	0.051860	-0.091464	0.185882	46	5000	12345	percentile

variable	effect	estimate	bootstrap_mean	ci_low	ci_high	n_complete_cases	n_bootstrap	seed	ci_method
percent_>=HighSch	ADE	-0.081172	-0.068740	-0.232951	0.127379	46	5000	12345	percentile
percent_>=HighSch	Total	-0.019858	-0.016879	-0.147985	0.123467	46	5000	12345	percentile

Across SES variables, the percentile bootstrap CIs for ACME all crossed zero (e.g., `Med_income` ACME = 0.000029, 95% CI: −0.000039, 0.000091), whereas `Med_income`'s ADE and total effects were negative with CIs excluding zero (ADE 95% CI: −0.000182, −0.000018; Total 95% CI: −0.000131, −0.000017). For `percent_Poverty`, `percent_Unemployed`, `percent_Uninsured`, and `percent_>=HighSch`, both ACME and ADE intervals spanned zero, and the corresponding total-effect intervals also spanned zero.

3.4.2 Spatial dependence and spatially varying associations

Spatial autocorrelation in outcome-model residuals was evaluated using the aligned Queen contiguity weights described in the workflow outputs. Table 6 reports the Global Moran's I for the residuals from the depression outcome model.

Table 6. Global Moran's I test (with permutation p-value) for spatial autocorrelation in Depression_prev outcome-model residuals using aligned Queen contiguity weights.

variable	morans_I	expected_I	variance_norm	z_norm	p_perm	permutations	n	s0	n_islands
outcome_residual	-0.067247	-0.022222	0.008419	-0.490694	0.321	999	46	46.0	0

Residual Moran's I was −0.067 (permutation p = 0.321; 999 permutations), indicating no evidence of positive spatial clustering in the modeled residuals under the specified weights.

A spatial error model (SEM) was also estimated for depression prevalence with the same predictor set and aligned Queen weights. Table 7 reports SEM coefficient estimates, the spatial error parameter λ, and model-fit statistics.

Table 7. Spatial error (SEM) regression results for Depression prevalence with aligned Queen contiguity weights (coefficients, lambda, standard errors, and fit statistics).

category	term	estimate	std_err	z	p_value
coefficient	CONSTANT	58.411955066529345	11.917606	4.901316	9.519672e-07
coefficient	Med_income	-9.633172785577203e-05	0.000037	-2.593223	9.508109e-03
coefficient	percent_Poverty	0.03801344179357413	0.062717	0.606107	5.444438e-01
coefficient	percent_Unemployed	-0.1129639237560518	0.153228	-0.737226	4.609849e-01
coefficient	percent_Uninsured	-0.12785191717234046	0.092621	-1.380372	1.674723e-01
coefficient	percent_>=HighSch	-0.07418646333801937	0.076455	-0.970324	3.318851e-01
coefficient	mobility_PC1	-0.1354727576226118	0.212390	-0.637850	5.235715e-01
coefficient	percent_>=18	-0.27210915834565963	0.060635	-4.487676	7.200444e-06
coefficient	percent_Black	-0.10917180560982409	0.010192	-10.711413	8.997787e-27
coefficient	percent_Hispanic	-0.11541303280259996	0.050091	-2.304077	2.121835e-02
coefficient	lambda	-0.2855481489441874	0.237603	-1.201786	2.294463e-01
model_fit	n	46	NaN	NaN	NaN
model_fit	k	10	NaN	NaN	NaN
model_fit	log_likelihood	-54.58391756956071	NaN	NaN	NaN

category	term	estimate	std_err	z	p_value
model_fit	aic	129.16783513912142	NaN	NaN	NaN
model_fit	bic_schwarz	147.45424910401238	NaN	NaN	NaN
model_fit	pseudo_r2	0.7352404143045579	NaN	NaN	NaN
model_fit	sigma2	[[0.61828934]]	NaN	NaN	NaN
model_fit	lambda	-0.2855481489441874	NaN	NaN	NaN

The SEM estimated $\lambda = -0.286$ (SE = 0.238, p = 0.229). Coefficient patterns broadly aligned with the non-spatial outcome model in sign for several predictors (e.g., `Med_income`: estimate = −9.63e−05, p = 0.00951; `percent_>=18`: estimate = −0.272, p < 0.001; `percent_Black`: estimate = −0.109, p < 0.001; `percent_Hispanic`: estimate = −0.115, p = 0.0212), while `mobility_PC1` was not estimated as non-zero (estimate = −0.135, p = 0.524).

Finally, a geographically weighted regression (GWR) was estimated for `Depression_prev` with a bisquare kernel and adaptive bandwidth selected by AICc (as described in the GWR outputs). Model diagnostics are provided in Table 8.

Table 8. Gwr model diagnostics.

n	y	predictors	kernel	fixed	spherical	bandwidth	bandwidth_criterion	aicc	aic	bic	R2	adj_R2	sigma2	scale
46	Depression_prev	Med_income, percent_	bisquare	False	False	43	AICc	115.596331	81.649385	118.358832	0.855677	0.749491	0.246565	0.246565

n	y	predictors	kernel	fixed	spherical	bandwidth	bandwidth_criterion	aicc	aic	bic	R2	adj_R2	sigma2	scale
		Poverty, percent_Unemployed, percent_Uninsured, percent_>=HighSch, mobilit												

n	y	predictors	kernel	fixed	spherical	bandwidth	bandwidth_criterion	aicc	aic	bic	R2	adj_R2	sigma2	scale
		y_PC1, percent_>=18, percent_Black, percent_Hispanic												

The GWR used an adaptive bandwidth of 43 (nearest neighbors), and the model-level fit statistics were R^2 = 0.856 and adjusted R^2 = 0.749 (AICc = 115.6). Spatial variability in local coefficient estimates is summarized in Table 9.

Table 9. Gwr coefficient spatial variability summary.

coefficient	mean	std	min	p25	median	p75	max	iqr
b_Intercep	51.428125	5.503089	40.158497	47.617743	51.136164	55.929938	60.660196	8.312195

coefficient	mean	std	min	p25	median	p75	max	iqr
b_Med_income	-0.000087	0.000031	-0.000140	-0.000113	-0.000082	-0.000061	-0.000036	0.000052
b_mobility_PC1	-0.246936	0.204731	-0.544388	-0.383671	-0.274737	-0.124070	0.177564	0.259601
b_percent_Black	-0.090640	0.012245	-0.108510	-0.101996	-0.091062	-0.081290	-0.068835	0.020705
b_percent_Hispanic	-0.107183	0.048379	-0.161001	-0.139187	-0.128122	-0.089011	0.014937	0.050176
b_percent_Poverty	0.045772	0.065634	-0.036841	-0.013886	0.027098	0.108347	0.167027	0.122234
b_percent_Unemployed	-0.177440	0.082330	-0.339567	-0.239304	-0.169253	-0.105459	-0.057456	0.133845
b_percent_Uninsured	-0.049257	0.085048	-0.229957	-0.101451	-0.022104	0.003864	0.074059	0.105315
b_percent__18	-0.234689	0.033570	-0.294623	-0.255331	-0.244844	-0.223773	-0.148703	0.031558
b_percent__HighSch	-0.048248	0.025351	-0.086949	-0.068111	-0.053828	-0.031427	0.017451	0.036684

Several predictors exhibited substantial local range in GWR coefficients, including `mobility_PC1` (min = −0.544, max = 0.178) and `percent_Poverty` (min = −0.0368, max = 0.167), while others were consistently negative across space (e.g., `Med_income` local coefficients remained negative throughout: min = −0.000140, max = −0.000036). Figure 4 maps the local GWR coefficients (centroid-based) for visual inspection of where coefficient magnitudes varied most.

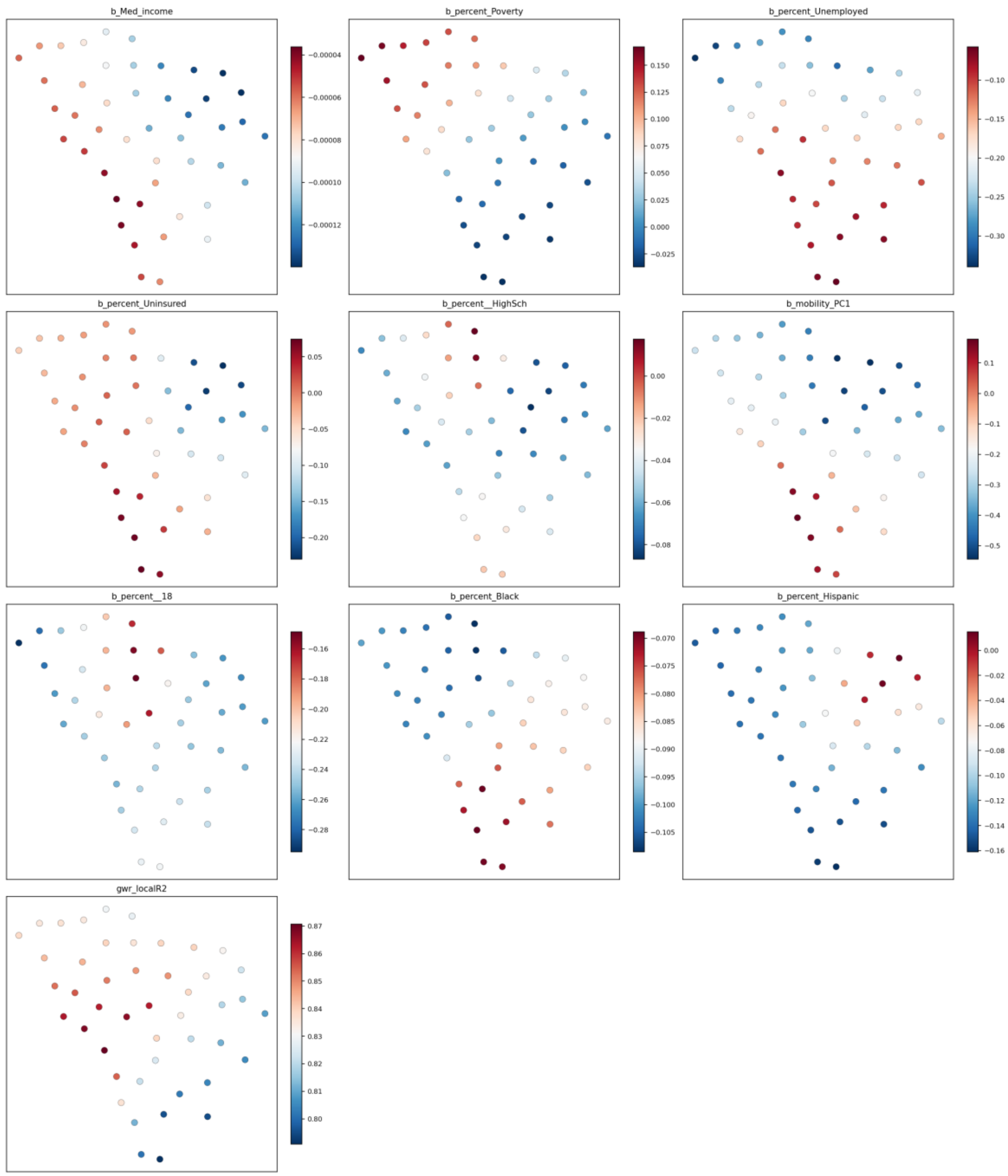

Figure 4: Gwr local coefficients centroid maps

The coefficient maps highlight that some covariates (notably `mobility_PC1`, `percent_Poverty`, and `percent_Uninsured`) exhibited pronounced geographic variation in both magnitude and—where ranges crossed zero—direction, whereas others (e.g., `Med_income`, `percent_Black`, and `percent_>=18`) were visually more uniform in sign across the state.

Hypothesis Assessment (Objective 3). The hypothesis stated that mobility composition (`mobility_PC1` and/or `mobility_diversity`) would transmit indirect effects from SES variables to depression prevalence and that accounting for spatial dependence would change inference on socioeconomic and mobility effects. For mediation, bootstrap ACME intervals for all SES variables crossed zero (e.g., `Med_income` ACME 95% CI: −0.000039, 0.000091), and `mobility_PC1` was not estimated as non-zero in the depression outcome model (β = −0.220, p = 0.321). For spatial dependence, residual Moran's I was not estimated as non-zero (I = −0.067, p = 0.321) and the SEM spatial parameter was not estimated as non-zero ($\lambda = -0.286$, p = 0.229), while key covariates (e.g., `Med_income`) remained negative in both OLS and SEM. However, GWR diagnostics and coefficient ranges indicated spatial variability in several local associations (e.g., `mobility_PC1` range: −0.544 to 0.178). Taken together, the mediation component was **not supported**, while the spatial-nonstationarity component was **partially supported** (via spatially varying GWR coefficients, despite weak evidence for residual spatial autocorrelation or a non-zero SEM λ).

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4. Discussion

4.1 Principal Findings

Across South Carolina counties, socioeconomic advantage and depression prevalence were systematically patterned, with higher county SES corresponding to lower estimated depression prevalence even after demographic adjustment. Mobility composition, summarized by a principal-component gradient of POI visitation, also co-varied strongly with SES, indicating that county socioeconomic structure was reflected in distinctive place-visitation profiles. However, the evidence did not support the specific pathway that mobility composition (as captured by `mobility_PC1`) statistically mediated the SES–depression relationship in this ecological setting: the mediator–outcome association was imprecisely estimated, and indirect effects (ACME) were not distinguishable from zero across SES indicators. Spatial diagnostics further suggested that residual spatial dependence was not a dominant feature of the global outcome model, while exploratory geographically weighted regression (GWR) summaries indicated meaningful spatial nonstationarity in several covariate associations, including the mobility term. The remainder of this Discussion interprets these patterns, revisits the preregistered hypotheses, connects results to the stated research gap, and outlines implications, limitations, and future research.

4.2 Interpretation

4.2.1 Socioeconomic gradients in depression prevalence: evidence consistent with contextual disadvantage mechanisms, but with ecological boundary conditions

Pattern. The county-level regression results were consistent with a socioeconomic gradient in depression prevalence: higher composite SES was associated with lower depression prevalence in the demographic-adjusted model (Table 1). This directly aligned with the direction of the preregistered **H1** expectation (lower SES ↔ higher depression prevalence). At the same time, several demographic composition variables were also negatively associated with depression prevalence in the same global specification, underscoring that county-level depression estimates co-varied with multiple dimensions of population structure rather than SES alone.

Mechanism. A plausible interpretation is that the observed SES gradient reflects a bundle of contextual processes—material conditions, stress exposures, access to services, and social-interactive neighborhood characteristics—that tend to cluster geographically and co-produce mental health disparities. This interpretation is consistent with "black box" accounts in which neighborhood socioeconomic conditions operate through intermediate social and interactional features (e.g., cohesion, trust, interaction opportunities) that shape psychosocial stress and mental health (Jakobsen et al., 2022), and with broader social-determinants arguments that mental health burdens are unequally distributed across social and spatial contexts (Satcher, 2001). Importantly, these mechanisms have clear **boundary conditions** in the present study: county-scale associations are most likely to approximate contextual mechanisms when (i) within-county heterogeneity is not overwhelming, (ii) county residence aligns reasonably with day-to-day social environments, and (iii) the depression prevalence estimates and SES measures refer to comparable time windows. Where counties contain strong internal segregation (urban–suburban–rural mixes) or where individuals' "effective neighborhoods" extend beyond county boundaries, county averages may not represent the operative contexts, weakening any link between county SES and mental health mechanisms that operate at finer scales.

Engagement with prior literature. The direction of the SES–depression association converges with the long-standing view that socioeconomic inequality is a fundamental correlate of health outcomes (Satcher, 2001) and with more recent attempts to specify intermediate contextual mechanisms behind SES–mental health relationships (Jakobsen et al., 2022). At the same time, the present study's design differs from much of that literature in two ways that likely shape interpretation. First, the analysis is ecological (county-level), whereas many mechanism-focused studies operate at neighborhood or individual levels and often incorporate survey-based mediators (e.g., cohesion and trust) (Oh and Thomas, 2024). Second, the outcome is an area-level modeled prevalence estimate rather than individually observed depression diagnoses. These differences mean that the present results are best read as **place-patterning evidence**—consistent with contextual disadvantage processes—rather than direct evidence that specific individuals experience SES-driven causal pathways.

Study-specific qualification. The negative associations of depression prevalence with county racial/ethnic composition in the global model (Table 1; also Tables 10–11) are not interpreted here as protective causal effects of race/ethnicity. At county scale, such coefficients can reflect reporting and diagnosis differences, age structure not fully captured by a single adult-share covariate, differential sampling/measurement properties embedded in modeled surveillance estimates, or unmeasured contextual correlates (e.g., urbanization, health care access, stigma). This is a core ecological inference limitation: coefficients can reflect compositional and measurement processes that differ from individual-level relationships (Coman et al., 2021).

Hypothesis revisit (H1). H1 was supported in the sense that the composite SES index—constructed from the preregistered SES components—was associated with lower depression prevalence in the expected direction after adjustment (Table 1). This support is conditional on the study's ecological scale and observational design: the results establish a robust county-level gradient but do not identify a unique causal effect of SES.

4.2.2 SES and mobility composition: strong gradients in type of visitation, weaker evidence for overall diversity differences

Pattern. The mobility models distinguished two different mobility constructs. The PCA-derived composition gradient (`mobility_PC1`) varied strongly with SES-related covariates (Table 2; Table 3), indicating that counties differed systematically in their visitation profiles by socioeconomic context. In contrast, mobility diversity (Shannon entropy across POI categories) showed comparatively weak and statistically uncertain relationships with the SES measures, with only limited evidence of association at conventional thresholds (Table 2).

Mechanism. A plausible mechanism is that SES shapes *opportunity structures* and constraints that affect **which kinds of destinations** are reachable, affordable, and routinely visited, more than it shapes the **evenness** of destination-category mixing. Counties with different economic bases, retail/service ecologies, transportation options, and spatial layouts may generate distinct visitation compositions (e.g., reliance on convenience-oriented retail versus recreation-oriented destinations) even if the overall number of categories represented in aggregate trips remains similar. This interpretation is consistent with the idea that passively collected, aggregated mobility data can reveal community-scale disparities in access and activity patterns (Lee et al., 2022). The **boundary conditions** are important: this mechanism would be expected to be weaker (i) if the mobility data disproportionately represent a subset of the population (e.g., smartphone users with consistent location services), (ii) if counties have similar POI supply mixes despite SES differences (e.g., regional chains homogenizing retail landscapes), or (iii) if cross-county travel is common such that county-of-residence does not align with county-of-visit, blurring the link between resident SES and observed destination composition.

Engagement with prior literature. The strong SES gradient in composition aligns with the "community-scale big data" argument that aggregated mobility traces can capture unequal interaction with place-based infrastructures (Lee et al., 2022). The weaker evidence for SES gradients in entropy-based diversity is not necessarily contradictory to mobility-inequality perspectives; rather, it suggests that **composition and diversity capture different aspects of mobility inequality**. Diversity indices can be insensitive when many categories are visited everywhere (raising baseline entropy) or when changes occur primarily through shifts in a few dominant categories while leaving the long tail of categories present (small entropy changes). Moreover, entropy measured across a fixed set of ten POI categories may not reflect meaningful diversity of *experiences* (e.g., quality, cost, social meaning) within categories—an important construct-validity limitation for interpreting “mobility diversity” as a proxy for opportunity.

Unexpected pattern and qualification. An unexpected element relative to **H2** was that diversity was not clearly lower in lower-SES counties, despite strong SES–composition links. Two study-specific considerations may account for this. First, the entropy measure depends on the distribution across broad categories; counties could differ markedly in the *dominant* categories (captured by PCA) while still exhibiting similar overall evenness. Second, the county scale may average over subcounty segregation: a county could contain both high-diversity urban centers and low-diversity rural areas, yielding moderate county entropy even when opportunity is unequally distributed internally. These issues constrain how directly the entropy results can be mapped onto “mobility opportunity” narratives without finer-scale analyses.

Hypothesis revisit (H2). H2 was partially supported. The “systematically different visitation composition” component was supported by the strong associations between SES indicators and the PCA-based composition gradient (Table 2). The “lower mobility diversity with lower SES” component was not supported at conventional thresholds in the entropy model (Table 2), suggesting that any SES–diversity relationship—if present—was smaller than could be resolved with 46 counties and the available measurement.

4.2.3 Mobility as a mediating pathway: why compositional mobility did not emerge as an indirect effect in county-level mediation

Pattern. Although SES covariates were strongly associated with `mobility_PC1` in the mediator model (Table 3), `mobility_PC1` was not precisely associated with depression prevalence in the outcome model that included SES and covariates (Table 4), and the bootstrapped ACME estimates for the hypothesized indirect pathway via `mobility_PC1` had percentile confidence intervals spanning zero across SES variables (Table 5). In other words, the data supported an SES → mobility-composition link, but did not support a mobility-composition → depression link sufficiently strong (conditional on SES and demographics) to yield evidence of mediation in this ecological specification.

Mechanism. Several non-exclusive mechanisms can explain this pattern without contradicting the broader premise that mobility environments matter for mental health. First, `mobility_PC1` may largely capture **structural land-use and retail ecosystem differences** (what kinds of POIs exist and are frequented) rather than psychosocial or therapeutic dimensions of mobility relevant to depression. Second, depression prevalence at county scale may be shaped more by **health care access, chronic stressors, and socioeconomic insecurity** than by destination-category composition per se; such upstream factors could dominate variation in depression prevalence even if mobility composition differs by SES. Third, the mediator could be misaligned with the true behavioral mechanism: for mental health, **social connectedness and supportive interactions**—not simply visiting certain POI categories—may be the operative pathway

(Jakobsen et al., 2022). Mobility composition could be an imperfect proxy for those interactional opportunities, especially when measured as aggregate county visitation counts rather than individual activity spaces and social encounters. The **boundary conditions** for a detectable mediation effect are therefore stringent: mediation would be more likely to emerge where (i) the mediator better captures psychosocially relevant mobility (e.g., third places, social venues, green-space use), (ii) temporal ordering is established (SES precedes mobility patterns, which precede depression changes), and (iii) confounding of mediator–outcome and exposure–mediator relationships is adequately controlled—conditions difficult to meet in cross-sectional, ecological county data.

Engagement with prior literature. The absence of detectable mediation via mobility composition contrasts with mediation studies that identify intermediate pathways from socioeconomic hardship to depression through explicitly psychosocial mediators (e.g., social cohesion, trust) (Oh and Thomas, 2024) or through modifiable health behaviors and conditions (Zhang et al., 2024). The divergence is plausibly attributable to differences in (i) mediator construct (mobility composition vs. psychosocial/behavioral health mediators), (ii) scale (county aggregates vs. individual measures), and (iii) design (cross-sectional ecological models vs. individual-level mediation frameworks with richer covariates). Rather than refuting the relevance of mobility to mental health, the present results suggest that **county-level POI composition summaries may be too distal—or too coarsely measured—to serve as a mediating construct** between SES and depression prevalence in this setting.

Study-specific qualification. Mediation analysis in observational settings is sensitive to unmeasured confounding and to model specification, particularly when both mediator and outcome are jointly influenced by unobserved features such as urbanization, service availability, and health care systems (Coman et al., 2021). As a result, the null ACME findings should be interpreted as "no evidence of mediation under this operationalization and scale," not as "evidence of no mobility pathway."

4.3 Closing the Loop with the Research Gap

The Introduction identified a gap in empirical work that links area-level SES to depression through **measurable, spatially explicit behavioral indicators** rather than relying primarily on survey-based contextual constructs or individual-level mediators, and it highlighted limited evidence on how mobility-derived measures can be leveraged to interrogate socioeconomic gradients in depression across local jurisdictions (Jakobsen et al., 2022); (Lee et al., 2022); (Coman et al., 2021). This study advanced that agenda by integrating county SES measures, PLACES depression prevalence, and multi-category POI visitation into a unified GIS workflow and by empirically demonstrating that (i) a county SES gradient in depression prevalence is detectable in South Carolina and (ii) SES differences are strongly reflected in mobility *composition*—a behavioral indicator that can be mapped and modeled. At the same time, the study clarifies what remains unresolved: **mobility composition as operationalized here did not function as a mediating pathway** in county-level mediation, implying that future work must refine mobility constructs, scale, and identification strategies if the goal is to open the “black box” with behavioral mobility indicators rather than psychosocial survey mediators.

4.4 Implications

A first implication is conceptual and methodological: mobility-based indicators appear valuable for characterizing how socioeconomic structure is expressed in **everyday place interaction patterns**, but different mobility summaries answer different questions. The strong SES associations for the PCA composition gradient indicate that POI visitation profiles can serve as a high-level “mobility environment signature” of county socioeconomic context (Lee et al., 2022). In contrast, entropy-based diversity provided limited evidence of SES patterning, suggesting that diversity indices over broad categories may be a weak proxy for opportunity differences at county scale. For applied GIScience, this implies that compositional approaches (e.g., PCA on category shares) may be more sensitive than single-number diversity metrics for detecting socioeconomic patterning in mobility systems, especially when the number of POI categories is modest.

A second implication concerns how mobility should be used in spatial health inference. The lack of mediation evidence suggests that researchers should be cautious about treating readily available mobility-composition measures as mechanistic intermediates between SES and mental health without validating the construct against psychosocial processes emphasized in mental health theory (Jakobsen et al., 2022). Where the aim is to evaluate plausible pathways, mobility indicators may need to be aligned more directly with hypothesized mechanisms (e.g., access to mental health services, green space use, social “third places,” or indicators of social isolation), potentially combining mobility with place attributes that better capture restorative or socially supportive environments.

A third implication is spatial: the residual Moran’s I test did not indicate strong leftover spatial autocorrelation (Table 6), and the spatial error model did not suggest a strong spatial error process via λ (Table 7), implying that the global covariate specification captured a substantial portion of broad spatial patterning in the depression outcome at the county level. Yet the GWR summaries indicated that several associations varied spatially (Tables 4–5; Figure 4), underscoring that “no strong global residual autocorrelation” does not imply “spatial stationarity.” For quantitative geographic analysis, this implies that spatial diagnostics should be paired with nonstationary exploratory tools when the substantive question concerns geographically varying relationships (Coman et al., 2021), while still recognizing that GWR is primarily descriptive and sensitive to bandwidth and multicollinearity.

4.5 Limitations and Validity

Several limitations qualify inference. First, the analysis was **cross-sectional and ecological**, using county averages and modeled prevalence estimates. This threatens **internal validity** for mediation and mechanism claims because temporal ordering cannot be established and because the mediator–outcome relationship may be confounded by unmeasured county characteristics (e.g., health care supply, diagnostic practices, urban–

rural structure). The most likely bias is that uncontrolled confounding inflates or attenuates associations unpredictably; in mediation settings, it often biases indirect effects toward zero when mediator measurement is noisy and confounding is present, potentially contributing to the null ACME findings.

Second, the operationalization of "mobility" relied on **aggregated POI visitation counts across ten broad categories**. This raises **construct validity** concerns: category composition may be only weakly related to psychosocial constructs (social support, isolation, restorative exposure) that are more directly tied to depression pathways (Jakobsen et al., 2022); (Oh and Thomas, 2024). Measurement error in the mediator (e.g., incomplete capture of visits, differential smartphone coverage by SES, or mismatch between resident county and visit location) would tend to attenuate mediator–outcome associations, biasing mediation estimates toward zero and making indirect effects harder to detect.

Third, the study's geographic unit (county) introduces **MAUP and spatial aggregation** concerns that threaten both **statistical-conclusion validity** (reduced power with $n = 46$) and **external validity** (generalizability to other states or to within-county neighborhood processes). County-level results may mask within-county inequities and may not transfer to urban regions where neighborhood scale better matches exposure and activity spaces. Relatedly, strong covariate correlations at county scale (e.g., income, education, insurance) can complicate interpretation of individual coefficients in multivariable models, potentially contributing to imprecision in the outcome model's mobility term and to instability across local (GWR) estimates.

Fourth, while spatial diagnostics suggested limited residual spatial autocorrelation in the global model (Table 6), this does not eliminate spatial dependence concerns entirely. Spatial structure can enter through spatially patterned omitted variables or through nonstationary processes; exploratory GWR variability (Tables 4–5) suggests that associations may differ across regions of the state. This raises a **model specification validity** issue: a single global model may be a poor approximation if relationships genuinely vary geographically, even when residual autocorrelation is modest.

What remains supported despite these limitations is a set of robust descriptive inferences at the county scale: (i) depression prevalence and SES were geographically heterogeneous (Figure 1), (ii) a SES gradient in depression prevalence was evident in adjusted global models (Table 1), and (iii) SES covariates strongly patterned mobility composition (Table 2), even if the hypothesized mediation via `mobility_PC1` was not supported (Table 5).

4.6 Future Research

Future work should pursue three directions tied directly to these limitations and unresolved findings. First, to address the construct-validity gap in the mediator, studies should develop **mechanism-aligned mobility measures**—for example, visits to green spaces or community-serving "third places," travel to mental health care providers, or mobility-based indicators of social isolation—and test whether these better explain depression gradients than broad POI-category composition. This would operationalize the contextual mechanisms emphasized in neighborhood mental health theory more directly (Jakobsen et al., 2022).

Second, to improve internal validity for pathway claims, future research should use **longitudinal or quasi-longitudinal designs**, leveraging repeated mobility observations and time-indexed depression estimates (or related mental health proxies) to establish temporal ordering and assess whether changes in mobility environments correspond to changes in mental health outcomes. Mediation frameworks are most informative when exposure, mediator, and outcome can be temporally sequenced (Oh and Thomas, 2024); (Zhang et al., 2024).

Third, to address MAUP and ecological inference constraints, subsequent analyses should replicate the workflow at **finer spatial units** (e.g., census tracts) or use **multilevel spatial models** that simultaneously represent within-county heterogeneity and between-county structure (Coman et al., 2021). This would help determine whether the strong SES–mobility composition relationship observed at county scale persists—and becomes more mechanistically interpretable—when mobility measures are aligned more closely with neighborhoods and daily activity spaces.

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5. Conclusion

Prior work has established that socioeconomic conditions and mental health burdens are geographically uneven, but it has remained unclear whether routine mobility—operationalized as the composition and diversity of place visitation—systematically tracks socioeconomic context in a way that helps account for county-level socioeconomic gradients in depression within a spatially explicit setting. For South Carolina, this study shows that (i) county socioeconomic advantage is reliably associated with lower depression prevalence, and (ii) socioeconomic context is strongly mirrored in *which* destination types comprise aggregate visitation (mobility composition), while (iii) this compositional stratification does not, in this ecological setting, translate into evidence that mobility composition statistically mediates the socioeconomic status–depression association. In other words, mobility composition appears to be an informative marker of place-based socioeconomic patterning, but not a supported explanatory pathway for the observed county depression gradient under the study's modeling assumptions.

At a higher level, the results suggest a spatially structured alignment between social disadvantage and both mental health burden and everyday activity environments, coupled with a conceptual separation between "composition" and "diversity" as mobility constructs: counties can differ markedly in visitation profiles without exhibiting comparably strong differences in entropy-based mixing across broad destination categories. The spatial diagnostics further indicate that the statewide, global depression model did not primarily fail due to strong residual spatial dependence, even as exploratory geographically weighted summaries were consistent with local variation in some associations, including the mobility term.

The contribution is bounded to an observational, county-level analysis of all 46 South Carolina counties, using area-level modeled depression prevalence, ACS-based socioeconomic and demographic composition, and aggregated visitation shares across a fixed set of POI categories; it does not support individual-level inference, causal claims, or mediation as a definitive mechanism. Hypothesis **H1 was supported** (lower SES co-varied with higher depression prevalence), **H2 was partially supported** (SES co-varied with mobility composition but not with lower mobility diversity), and the hypothesized mediation pathway via mobility composition was **not supported**. The next step enabled by these findings is to test whether the same SES–mobility composition relationship predicts depression gradients when measured at finer spatial scales or with temporally aligned mobility and mental health data that better capture within-county heterogeneity and local nonstationarity.

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